Wild Quest

2D Game Document

Thilakarathna H.M.P.M.K.

IT22035394

Table of Contents

1. Game Concept 3

2. Core Features Implementation 3

2.1. Player Movement 3
2.2. Enemy Detection and Movement 4
2.3. Combat Mechanics 4
2.4. Player Health System 4
2.5. Timer System 5
2.6. Coin Collection System 5
2.7. Scene and Level Management 5
2.8. UI System 6

3. Screenshots of Gameplay 8

4. Control Guide 11

5. Creative Features 11

6. Credits 12

1. Game Concept

Wild Quest is a 2D platformer game where the player traveling through different forest environments. The main objective is to collect coins and treasures while navigating through levels and avoiding or defeating enemies.

The game includes multiple levels with increasing difficulty. Each level introduces environmental variation and enemy interactions. A timer system adds controlled pressure during enemy encounters, requiring the player to act within a limited time.

The overall gameplay is designed to be simple, smooth, and engaging while maintaining a calm and natural theme.

2. Core Features Implementation

2.1. Player Movement

- The player movement is implemented using Unity's Rigidbody2D component to ensure smooth and physics-based motion.
- Horizontal movement controlled using keyboard input. This allows smooth acceleration and deceleration.
- The velocity of the player is updated continuously by modifying the Rigidbody2D's linear velocity.
- Jumping is enabled only when the player is grounded using a ground check system and it is implemented using a force applied on the Y-axis when the player presses the spacebar.
- Player direction is handled using sprite flipping.

2.2. Enemy Detection and Movement

- Enemies detect the player within a defined distance.
- Each enemy has a health system requiring multiple hits to be defeated. This system allows progressive damage instead of instant elimination.
- The distance between the player and enemy is calculated. When the player enters a defined detection range, the enemy starts moving toward the player.
- Enemy direction is handled by flipping the sprite based on the player's position.

2.3. Combat Mechanics

Enemies can be defeated using two different mechanics implemented in the game.

1. Platforming Mechanic

Player can defeat enemies by jumping on top of them. A collision check is used to determine whether the player is above the enemy and moving downward. If this condition is satisfied, damage is applied to the enemy and the player performs a small bounce effect. This bounce prevents repeated automatic hits and improves gameplay control.

2. Projectile Mechanic

Player can use a projectile attack. When the player presses the right mouse button, the player shoots a bullet projectile in the direction the player is facing. The projectile reduces the enemy's health when it hits. After the enemy receives three successful hits, the enemy is destroyed.

This dual combat approach allows the player more interactions and flexibility during gameplay.

2.4. Player Health System

- The player has multiple lives displayed using a heart UI system. When damage occurs the life count decreases and the UI updates accordingly

- Player can lose lives when:
  - Collides incorrectly with an enemy
  - Timer conditions are triggered
  - Player falls into a death zone (water/spikes)
- Health packs are placed throughout the levels to increase the player's lives.

2.5. Timer System

- A conditional timer system is used during enemy encounters. The timer starts only when the player enters the enemy's detection range.
- The countdown runs from 15 seconds. Life reduction conditions:
  - Lose one life at 10 seconds
  - Lose another life at 5 seconds
  - Final life lost at 0 seconds
- The timer is displayed on UI during combat and stopped when the enemy is defeated or the player lost the game.
- This timer system adds controlled difficulty without affecting the entire level.

2.6. Coin Collection System

- Coins are placed across levels as collectible objects. Player collects coins by collision and the game score increases accordingly.
- This encourages exploration and adds a reward system.

2.7. Scene and Level Management

- The game uses multiple scenes to represent levels and the scene transitions are handled using Unity's Scene Manager.
- The game consists of three different levels, each designed with a different environmental theme to create variety in gameplay and visual appearance.

- **Level 1 – Spring Environment**
  The first level introduces the player to the game mechanics. The environment is designed with a spring theme, using bright colors, green landscapes, and natural elements. In this level the player learns how to move, jump, collect coins, and interact with enemies.

- **Level 2 – Winter Environment**
  The second level changes the environment to a winter theme. Snow covered platforms, colder color tones, and winter scenery are used to create a different visual experience. The level contains additional enemies and obstacles, increasing the challenge compared to the first level.

- **Level 3 – Autumn Environment**
  The third level uses an autumn themed environment, featuring warm colors such as orange and brown to represent falling leaves and seasonal change. This level is designed to be the most challenging stage, combining enemies, obstacles, and coin collection elements.

- Scene management is handled through Unity's scene loading system, ensuring smooth progression from one level to the next.
- Restart functionality reloads the current scene and resets player state, lives, and UIs.

2.8. UI System

- The game includes several UI elements that provide feedback to the player and control different stages of the game. These UI components are created using Unity's Canvas system and are activated or deactivated through scripts based on gameplay events.
  - **Start Menu**

  The Start Menu appears when the game is launched. It allows the player to begin the game and access the main interface before entering the first level. This screen acts as the entry point to the game.

- **Timer Panel**

The Timer Panel displays a countdown timer when the player encounters an enemy. The player must defeat the enemy within the given time limit. If the timer reaches zero before the enemy is defeated, the player loses a life. This system adds pressure and encourages quick reactions.

- **Game Over Panel**

The Game Over Panel appears when the player loses all available lives. It informs the player that the game has ended and provides options such as restarting the level or returning to the menu.

- **Level Complete Panel**

When the player successfully finishes a level, the Level Complete Panel is displayed. This panel confirms that the level objectives are completed and allows the player to proceed to the next level.

- **Game Win Panel**

The Game Win Panel appears after completing the final level with the total score. It indicates that the player has successfully finished the entire game.

- **Settings Panel**

The Settings Panel allows the player to adjust certain game options, such as audio settings. This panel can be opened during the game and provides basic control over the game environment.

- **Chest Collected Panel**

When the player collects a treasure chest in the level, the Chest Collected Panel appears to notify the player of the successful collection. This UI element provides feedback and reinforces the reward system in the game.

3. Screenshots of Gameplay

Figure 1 - Main Menu

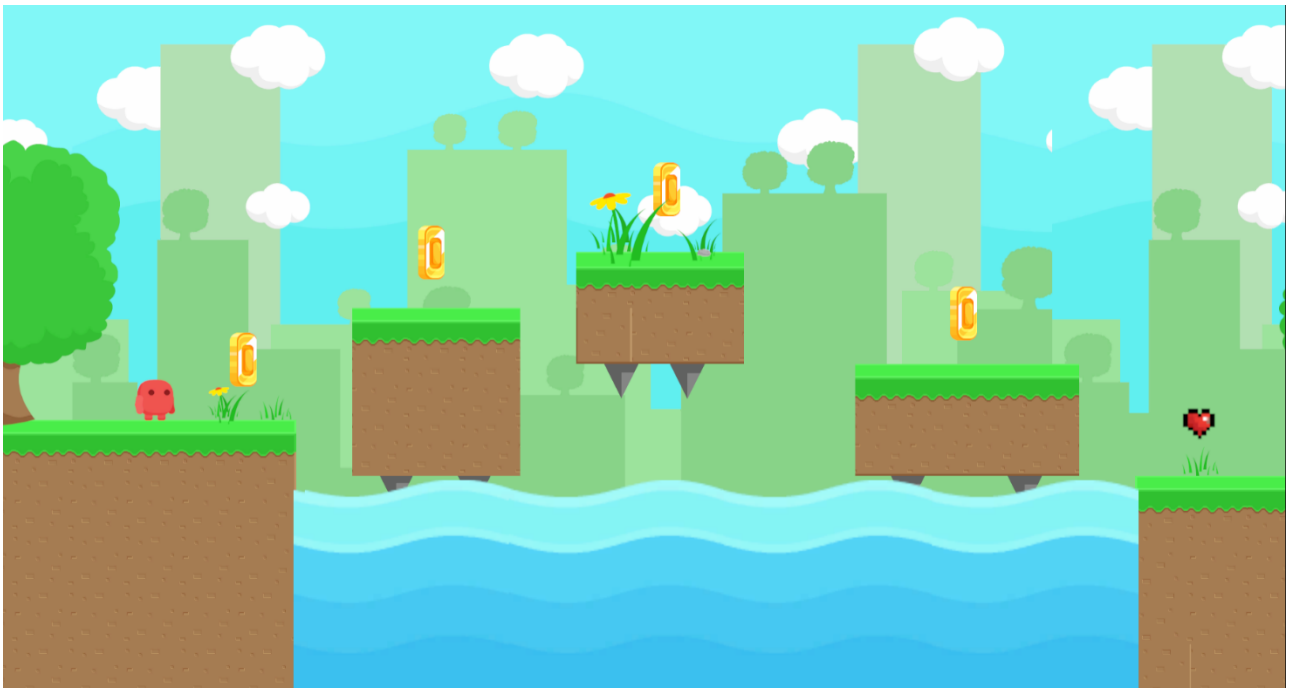

Figure 2 - Game Level 1

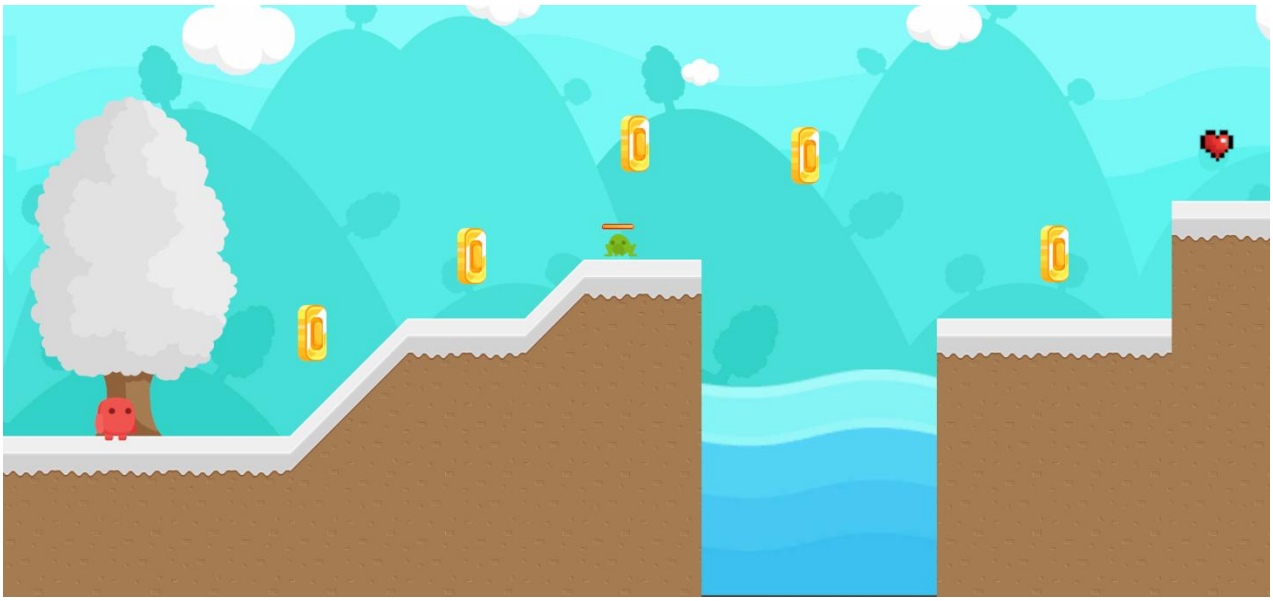

Figure 3 - Game Level 2

Figure 4 - Game Level 3

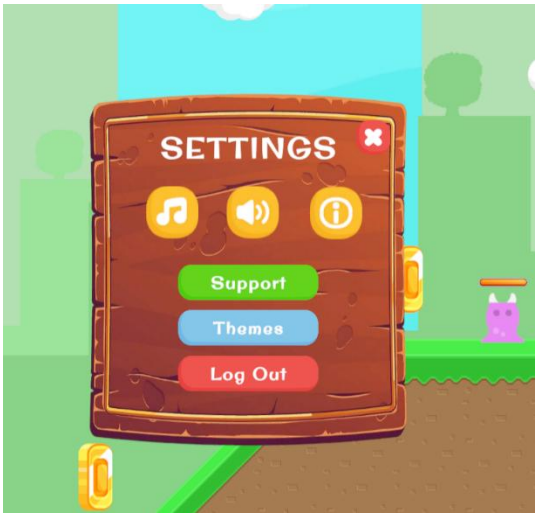

Figure 5 - Settings Panel

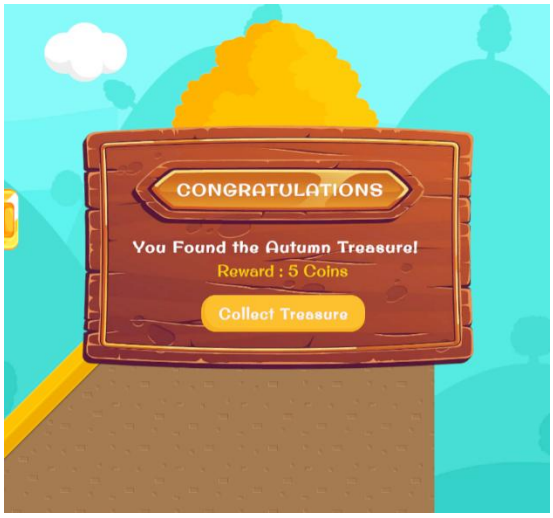

Figure 6 - Treasure Panel

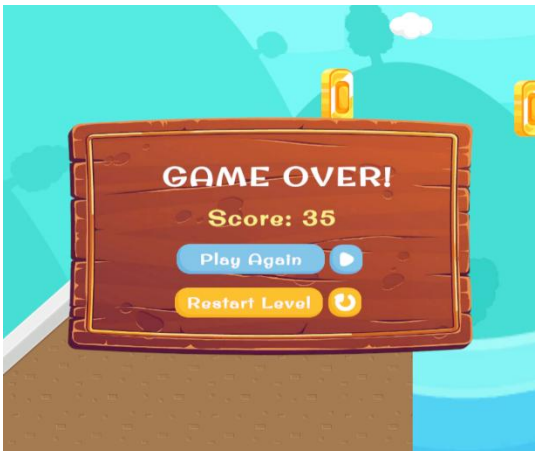

Figure 7 - Game Over Panel

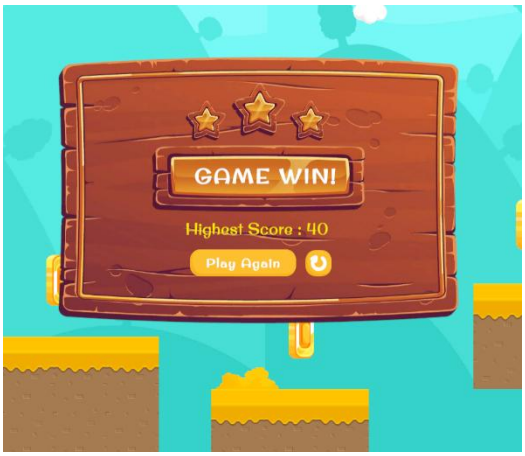

Figure 8 - Game Win Panel

Figure 9 - Level Complete Panel

Figure 10 - Instructions Panel

4. Control Guide

Action	Input
Move Left / Right	A / D or Arrow Keys
Jump	Spacebar
Shoot Projectile	Right Mouse Click
Pause / Settings	UI Button

5. Creative Features

A countdown timer activates when the player encounters an enemy. The player must defeat the enemy before the timer reaches zero. If the timer runs out, the player loses a life. This mechanic introduces a controlled challenge during combat without affecting the entire level. It encourages the player to react quickly and make strategic decisions when facing enemies.

Another creative element implemented in the game is the danger overlay system. When the player is reduced to the last remaining life, a red screen overlay appears on the display. This visual effect signals that the player is in a critical state and increases the sense of urgency. The overlay remains active until the player gains health or the game ends.

Additional creative elements included in the game are:

- Multiple enemy types such as frogs and snails, which add variation to enemy encounters.
- Health recovery items that allow the player to regain lost lives by collecting health packs during gameplay.

6. Credits

Game Engine	Unity Game Engine
Programming Language	C#
Assets	Unity Asset Library Kenney Assets Freepik Game Assets
Audio	Pixabay Sound Effects

Game Demonstration Video - Game Demo Video.mp4